

Do Robotic World Models Really Follow Actions?

Diagnosing and Aligning Action-Conditioned Generation for Policy Learning

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Abstract

Action-conditioned world models are increasingly used as learned simulators for policy evaluation and improvement, yet their effectiveness rests on an unverified assumption: generated futures faithfully reflect arbitrary valid actions. Existing benchmarks are typically confined to expert demonstrations, leaving off-expert action following inadequately evaluated. To address this gap, we introduce WorldEcho, which probes action following over a broader action distribution using visual integrity and SE(3) trajectory alignment. Our diagnosis shows that current world models reasonably execute expert actions but struggle with diverse off-expert trajectories, either ignoring the commanded actions or producing visually invalid roll-outs. We further propose WorldSync, which strengthens action following along three complementary axes: distributional coverage, representational grounding, and intervention-effect alignment. It broadens the training distribution over action consequences, grounds intermediate video representations in action-induced robot dynamics through an Action-Forcing Expert, and aligns predicted changes under action interventions with the corresponding changes in ground-truth futures. Experiments on RoboTwin benchmarks and real-robot tasks show that WorldSync improves WorldEcho metrics and serves as a more reliable simulator for iterative policy improvement, enabling policies to achieve higher success rates.

1 Introduction

Recently, Vision-Language-Action (VLA) (Zitkovich et al. 2023; O’Neill et al. 2024; Black et al. 2025) and World Action Models (WAM) (Wu et al. 2024; Li et al. 2026a; Ye et al. 2026) have demonstrated promising success rates and generalization across diverse scenarios and tasks through large-scale pretraining. Despite these advances, prior works (Physical Intelligence et al. 2025; Pan et al. 2026; Guo et al. 2026a) have shown that task-specific online post-training remains essential for achieving optimal downstream performance. Such post-training, however, requires extensive interaction with real-world environments, which is costly and time-consuming (Xiao et al. 2026). To reduce this burden, action-conditioned world models (AC-WMs) (Zhu et al. 2025; Guo et al. 2026b) have been introduced as learned simulators to support efficient closed-loop policy interaction (Quevedo

et al. 2026; Li et al. 2025b). Subsequent studies have leveraged AC-WMs to provide synthetic experience for policy post-training (Guo et al. 2026a; Yu et al. 2026; Xiao et al. 2026; Li et al. 2025a; Jiang et al. 2026b), leading to improved policy performance in real-world deployment. Nevertheless, these approaches rest upon a critical yet largely unverified assumption: **AC-WMs genuinely capture world dynamics and produce accurate responses to arbitrary valid action inputs** (Quevedo et al. 2026; Guo et al. 2026a; Yang et al. 2026a).

Faithful action following is a prerequisite for using AC-WMs as reliable simulators for policy evaluation and post-training (Li et al. 2026b). For evaluation, plausible but action-inconsistent futures can misrepresent the consequences of candidate actions, creating a gap between simulated and real-world policy performance (Quevedo et al. 2026; Li et al. 2025b). For post-training, unreliable rollouts require additional verification, rejection, or filtering before they can provide trustworthy supervision (Guo et al. 2026a; Yu et al. 2026), increasing overhead and reducing the yield of usable experience. Improving action following can therefore enhance evaluation fidelity and deliver more useful training data under a fixed interaction and generation budget. Existing world model benchmarks mainly assess perceptual and semantic quality, similarity to reference behaviors, or downstream executability (Hu et al. 2025; Shang et al. 2026a; Jiang et al. 2026a). Recent studies have begun to examine out-of-distribution or failure-inducing actions (Quevedo et al. 2026; Yang et al. 2026a), but continuous action following across broad numerical queries with action-specific SE(3) ground truth remains underexplored. Such off-expert actions are essential for policy improvement because learned policies inevitably induce state-action distributions beyond expert demonstrations (Ross, Gordon, and Bagnell 2011; Yu et al. 2026; Guo et al. 2026a).

Figure 1 summarizes our diagnose-improve-validate workflow: WorldEcho probes AC-WMs with demonstrated and off-expert action queries, WorldSync targets the diagnosed visual and action-alignment errors, and downstream policy improvement evaluates the utility of the resulting model. To fill this evaluation gap, we introduce WorldEcho, which evaluates action following across five complementary action-query categories. In addition to demonstrated actions as an in-distribution baseline, we construct four off-

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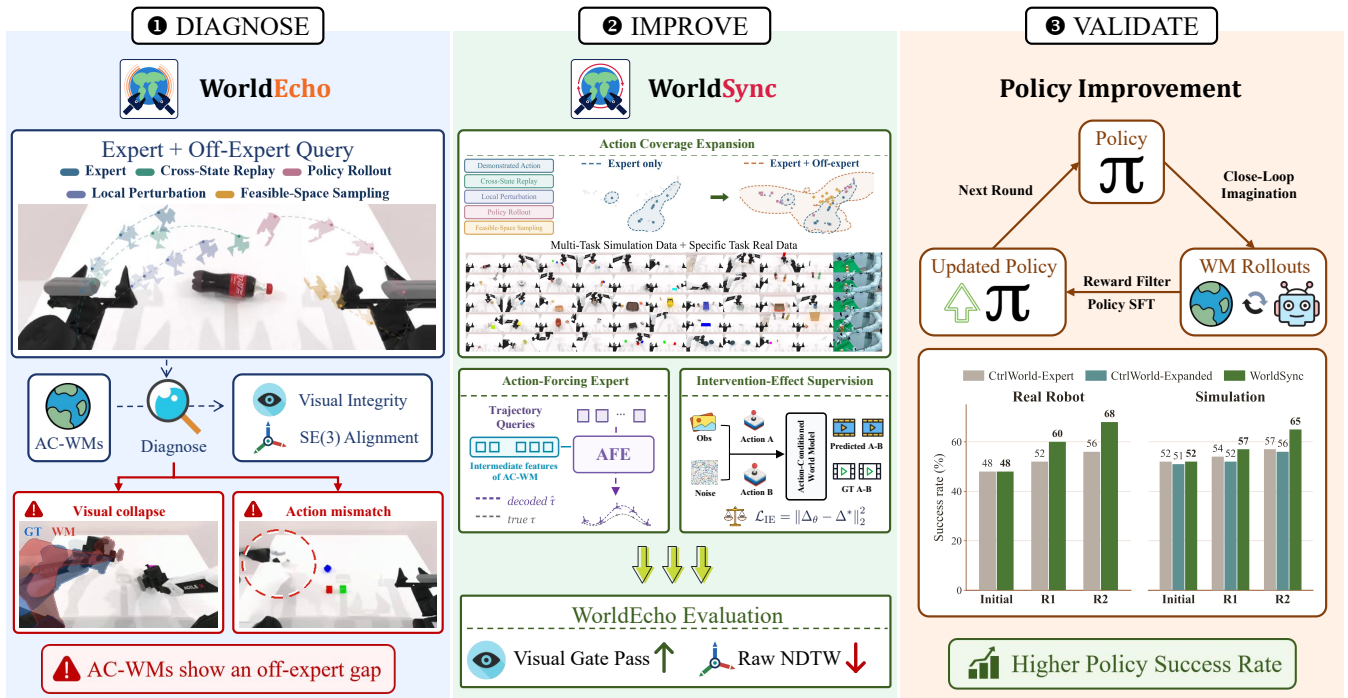


Figure 1: Overview of our diagnose-improve-validate pipeline. WorldEcho probes action-conditioned world models with demonstrated and diverse off-expert queries, jointly evaluating visual integrity and SE(3) end-effector alignment to expose visual collapse and action mismatch. Guided by this diagnosis, WorldSync broadens the training distribution over action consequences, grounds intermediate video representations in action-induced robot dynamics through an Action-Forcing Expert, and aligns predicted changes under action interventions with the corresponding changes in ground-truth futures. In simulation and real-robot policy-improvement experiments, these gains translate into higher policy success rates.

expert categories that progressively reduce their reliance on expert behavior: *Cross-State Replay*, *Local Perturbation*, *Policy Rollout*, and *Feasible-Space Sampling*. These categories respectively diagnose reliance on state-conditioned expert priors, sensitivity to local action variations, fidelity under policy-induced deviations, and controllability across the broader feasible action space. To capture distinct sources of simulation error, WorldEcho jointly evaluates the visual integrity of generated videos and the SE(3) alignment between end-effector trajectories extracted from generated and corresponding ground-truth videos. Our diagnosis reveals an off-expert support gap: expert-only AC-WMs follow demonstrated actions reasonably well but exhibit two characteristic failure modes under off-expert actions. They either produce visually plausible yet overly optimistic futures that deviate from the conditioned actions, or lose visual integrity through severe degradation, such as distorted robot arms and disappearing grippers. Together, these failures expose two limitations of expert-only AC-WMs: narrow support over off-expert action consequences and weak dependence of generated dynamics on the conditioned actions.

Guided by this diagnosis, we propose WorldSync, a systematic training recipe that strengthens action-conditioned generation along three complementary axes: distributional coverage, representational grounding, and intervention-effect alignment. First, to expand the training distribution over ac-

tion consequences, we unify diverse simulated expert and off-expert trajectories with a small amount of target-domain real-world data in a shared action space, broadening action support while preserving real-world visual fidelity. With this broader support established, we introduce an *Action-Forcing Expert* (AFE) that decodes future robot states from intermediate video representations, thereby grounding the learned features in the robot dynamics induced by the conditioned actions. Yet feature-level grounding supervises each rollout in isolation and does not explicitly constrain how predictions should change across actions. We thus introduce *Intervention-Effect* (IE) supervision, which uses paired trajectories with the same observation but different actions to align predicted changes under an action intervention with the corresponding changes in ground-truth futures. In short, coverage expansion broadens the action consequences from which the model learns, AFE grounds what its representations encode in robot dynamics, and IE aligns how its predictions change with how the ground-truth futures change.

Experiments on RoboTwin (Chen et al. 2026b) and real-robot tasks demonstrate that WorldSync improves WorldEcho performance across both demonstrated and off-expert action queries while maintaining visual integrity. More importantly, when used as a learned simulator for iterative policy improvement, WorldSync provides more reliable action-dependent feedback and enables policies to achieve higher

success rates, demonstrating the practical value of faithful action following.

Our contributions are as follows:

- We introduce WorldEcho, which evaluates action following across demonstrated actions and four complementary off-expert action-query categories through visual integrity and SE(3) trajectory alignment.
- We identify two characteristic failures of expert-only AC-WMs under off-expert actions: visually plausible but overly optimistic futures that disregard the conditioned actions, and severe visual degradation that renders the generated rollouts unusable.
- We propose WorldSync to broaden the training distribution over action consequences, ground video representations in action-induced robot dynamics, and align predicted changes under action interventions with their ground-truth counterparts, enabling more faithful simulation and more effective policy improvement.

2 Related Work

2.1 Action-Conditioned Robotic World Models

Action-conditioned robotic world models predict future observations from visual histories and continuous robot commands, supporting imagined rollouts for planning and policy learning (Zhu et al. 2025; Guo et al. 2026b). Building on this formulation, subsequent work has advanced AC-WMs through architectural innovations for stronger action control and long-horizon generation (Zhu et al. 2025; Guo et al. 2026b; Zheng et al. 2026), as well as broader pretraining for transferring interaction dynamics across tasks and embodiments (Gao et al. 2026; Wu and Gao 2026; Huang et al. 2026). Beyond improving architectures and data, several methods address the representation gap between numerical commands and pixel-space motion using spatially grounded representations of robot kinematics (Wu and Gao 2026; Chen et al. 2026c,a; Yang et al. 2026b). Controlled action perturbations and counterfactual behaviors have also begun to probe models beyond standard action replay (Gao et al. 2026; Huang et al. 2026; Feingold et al. 2026; Zhu et al. 2025). However, these studies cover limited action variations and provide mostly indirect or coarse-grained evidence of action following rather than ground-truth motion for each command (Zhu et al. 2025; Gao et al. 2026; Huang et al. 2026; Feingold et al. 2026; Yang et al. 2026a).

2.2 World Models for Policy Evaluation and Improvement

AC-WMs are increasingly used as policy-facing simulators that reduce costly real-robot interaction (Quevedo et al. 2026; Li et al. 2025b; Jeon et al. 2026; Ma et al. 2026). In this role, imagined rollouts support policy evaluation and ranking (Quevedo et al. 2026; Li et al. 2025b; Jeon et al. 2026; Ma et al. 2026). Beyond evaluation, they provide synthetic experience or optimization signals for policy improvement (Yu et al. 2026; Xiao et al. 2026; Li et al. 2025a; Jiang et al. 2026b). These applications, however, expose a distribution shift when policy exploration, failures, or updates depart

from expert-dominated AC-WM data (Jiang et al. 2026b; Yin et al. 2026; Liu et al. 2026; Yu et al. 2026). To mitigate this mismatch, existing systems expand training with exploratory or corrective interactions (Yin et al. 2026; Li et al. 2026b; Xiao et al. 2026), filter unreliable generations, or co-evolve the policy and world model (Jiang et al. 2026b; Guo et al. 2026a; Liu et al. 2026). While these strategies improve downstream policy performance, policy gains alone do not reveal whether the world model faithfully responds to the queried actions or merely serves as useful visual augmentation (Guo et al. 2026a; Jiang et al. 2026b; Xiao et al. 2026; Li et al. 2025a). This ambiguity calls for directly assessing whether policy-facing AC-WMs faithfully respond to the actions queried by the policy.

2.3 Robotic World Model Evaluation

Robotic world-model evaluation has expanded beyond generic video quality to motion, physical consistency, and embodied functionality (Hu et al. 2025; Shang et al. 2026a,b). Existing benchmarks compare generated motion with reference trajectories (Hu et al. 2025; Shang et al. 2026a) or recover actions from generated videos to test embodied executability (Jiang et al. 2026a; Fan et al. 2026). Most closely related, MiraBench evaluates action-conditioned reliability with failure-inducing perturbations and reveals that visually plausible predictions may ignore commanded failures or exhibit optimism bias (Yang et al. 2026a). However, these evaluations rely primarily on image-space motion, recovered-action execution, or task-level failure judgments rather than continuous end-effector motion in SE(3) across broad numerical action queries (Shang et al. 2026a; Jiang et al. 2026a; Yang et al. 2026a). Our benchmark instead aligns generated end-effector trajectories with the corresponding simulator-replayed trajectories in SE(3) across progressively broader numerical action queries.

3 Method

3.1 Problem Formulation

We consider an action-conditioned robotic world model, parameterized by θ , that generates future visual observations conditioned on the current observation, task instruction, and robot actions. Let o_0 denote the current multi-view observation, c the language instruction, and $a_{1:H} = (a_1, \dots, a_H)$ a sequence of numerical robot actions over a horizon of H steps. We model the future multi-view video $I_{1:H}$ with the conditional distribution

$$p_\theta(I_{1:H} | o_0, c, a_{1:H}), \quad \hat{I}_{1:H} \sim p_\theta(\cdot | o_0, c, a_{1:H}). \quad (1)$$

Executing the same action sequence from the corresponding initial environment state yields the ground-truth future $I_{1:H}^{\text{GT}}$. Let Φ extract the end-effector trajectory from a multi-view video. The generated and ground-truth trajectories are

$$\hat{\tau} = \Phi(\hat{I}_{1:H}), \quad \tau^{\text{GT}} = \Phi(I_{1:H}^{\text{GT}}). \quad (2)$$

A reliable world model should generate a visually valid future whose induced robot trajectory agrees with the ground-truth consequence of the queried actions. We study how to evaluate and improve this property when $a_{1:H}$ extends beyond the expert action distribution.

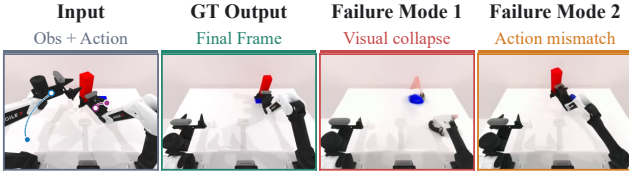


Figure 2: Action-following failures under off-expert actions. Compared with the ground truth, an expert-trained AC-WM either suffers visual collapse (Failure Mode 1) or generates plausible but action-inconsistent motion (Failure Mode 2).

3.2 Motivation

Most robotic world models are trained on expert demonstrations (Zhu et al. 2025; Guo et al. 2026b; NVIDIA et al. 2025). Yet policy evaluation and improvement inevitably query actions beyond the expert distribution (Quevedo et al. 2026; Li et al. 2025b; Yu et al. 2026; Xiao et al. 2026), requiring the model to respond faithfully to diverse valid actions. To examine whether existing models satisfy this requirement, we fine-tune Cosmos-Predict2.5 (NVIDIA et al. 2025) on expert demonstrations collected in the RoboTwin simulation benchmark (Mu et al. 2025; Chen et al. 2026b). Starting from the same observation, we condition it on either expert or feasible off-expert actions and compare its predictions with the ground-truth rollouts obtained by executing the queried actions in RoboTwin. The model accurately replays expert trajectories, but exhibits two distinct failures under off-expert actions, as shown in Figure 2: the generated video either loses visual integrity, with distorted arms or disappearing grippers, or remains visually plausible while depicting motion inconsistent with the queried actions. These observations motivate a benchmark that probes world models over a broader action distribution and jointly evaluates visual integrity and the fidelity of action-induced motion.

3.3 WorldEcho: Benchmarking Action Following

Motivated by the failures above, we introduce WorldEcho to evaluate whether an AC-WM faithfully responds to numerical robot actions beyond expert replay. As illustrated in Figure 3, WorldEcho expands the queried action distribution from demonstrated actions to four complementary off-expert categories. For every query, we execute the same action sequence from the corresponding initial state in RoboTwin to obtain a ground-truth future. We then evaluate the generated rollout from two complementary perspectives: whether it remains visually valid and whether its induced end-effector motion agrees with the ground-truth consequence of the queried actions.

Action Query Construction We construct five action-query categories with decreasing reliance on the joint expert distribution over states and actions. *Demonstrated Action* replays the expert action sequence associated with the current observation and serves as the in-distribution baseline. *Cross-State Replay* applies an expert action sequence from another state. Although the actions remain expert-distributed, their mismatch with the current state induces task failure, testing

whether the model captures state-dependent action effects rather than equating expert-like actions with success. *Local Perturbation* applies bounded perturbations to demonstrated actions, probing sensitivity to local changes around the expert manifold. *Policy Rollout* uses actions produced by a learned policy, reflecting the deviations encountered during policy evaluation and improvement. Finally, *Feasible-Space Sampling* draws valid actions from the broader robot action space to assess controllability with minimal reliance on expert behavior. All off-expert queries are filtered for action feasibility and replayed in RoboTwin from the same initial state as the world-model query, producing an action-specific ground-truth rollout.

Visual Integrity Assessment Trajectory agreement is meaningful only when the generated video remains a valid depiction of the robot and scene. We therefore assess four complementary aspects of visual integrity. Following WorldArena (Shang et al. 2026a), we adopt continuous scores for image quality and motion smoothness. Specifically, image quality q measures frame-level perceptual fidelity using MUSIQ (Ke et al. 2021), while motion smoothness m evaluates temporal continuity through frame-interpolation consistency (Zhang et al. 2024). To integrate these scores into our integrity-gated protocol, we convert them into binary decisions using prespecified thresholds:

$$G_{\text{quality}} = \mathbb{I}[q \geq \tau_q], \quad G_{\text{motion}} = \mathbb{I}[m \geq \tau_m]. \quad (3)$$

where $\mathbb{I}$ is the indicator function and τ_q, τ_m are the respective thresholds. End-effector visibility G_{EEF} and arm integrity G_{arm} directly produce binary decisions. The former tracks the grippers throughout the rollout using SAM-based video tracking (Carion et al. 2026); the latter detects blurred, broken, or disappearing robot arms using a vision-language evaluator (Bai et al. 2025). All criteria and thresholds are fixed across evaluated models. The overall visual-integrity gate is

$$G_{\text{vis}} = G_{\text{quality}} \wedge G_{\text{motion}} \wedge G_{\text{EEF}} \wedge G_{\text{arm}}. \quad (4)$$

A rollout passes the gate only when all four conditions are satisfied.

End-Effector Trajectory Alignment A visually valid rollout may nevertheless ignore the conditioned actions. We therefore directly compare the robot motion in the generated and ground-truth videos. Given a video I , the trajectory extractor Φ (Tan et al. 2026) recovers the per-frame position $p_t^e \in \mathbb{R}^3$ and orientation $R_t^e \in \text{SO}(3)$ for each end effector $e \in \{L, R\}$ (left or right). For a generated frame i and a ground-truth frame j , we define the local pose discrepancy as

$$d_e(i, j) = \left[w_p^2 \|\hat{p}_i^e - p_j^{e, \text{GT}}\|_2^2 + w_R^2 d_{\text{SO}(3)}^2(\hat{R}_i^e, R_j^{e, \text{GT}}) \right]^{1/2}, \quad (5)$$

where hatted and GT quantities are the generated and ground-truth poses, and w_p, w_R weight translation and rotation. The rotational discrepancy is

$$d_{\text{SO}(3)}(R_1, R_2) = \arccos(\text{clip}(\xi, -1, 1)), \quad \xi = \frac{\text{tr}(R_1^T R_2) - 1}{2}, \quad (6)$$

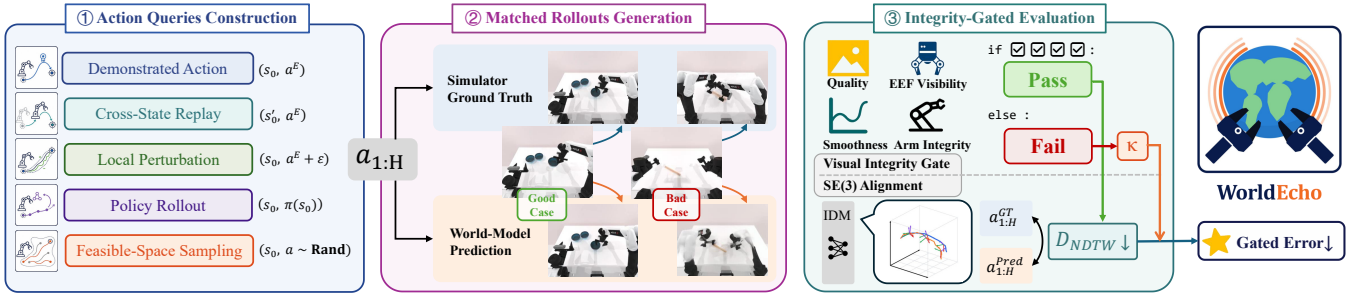


Figure 3: Overview of WorldEcho. Five action query categories span demonstrated and off-expert actions. Each action sequence produces matched simulator reference and world model rollouts. Visual integrity and SE(3) end effector trajectory alignment are jointly evaluated. The gated error uses NDTW for valid rollouts and a fixed penalty κ otherwise.

where tr is the matrix trace and clip clamps its argument to $[-1, 1]$. To accommodate differences in temporal progression, we align the two trajectories using pose-aware normalized dynamic time warping (NDTW) (Sakoe and Chiba 1978; Salvador and Chan 2007; Hu et al. 2025; Shang et al. 2026a). Let π_e denote the optimal warping path for end effector e . The sample-level NDTW error is

$$D_{\text{NDTW}} = \frac{1}{|\mathcal{A}|} \sum_{e \in \mathcal{A}} \frac{1}{|\pi_e|} \sum_{(i,j) \in \pi_e} d_e(i,j), \quad (7)$$

where $\mathcal{A}$ is the set of valid end effectors. We normalize the cumulative cost by alignment-path length, but not by the spatial extent of the reference trajectory. This choice retains the absolute metric scale of the pose discrepancy, preventing short-range motions from disproportionately amplifying pose-estimation noise and preserving a consistent physical interpretation across action queries. Lower values indicate stronger agreement between generated motion and the action-specific ground truth.

Integrity-Gated Evaluation Protocol For every query n , we retain both the visual-gate result G_n and the ungated NDTW error D_n^{NDTW} . We compute NDTW for all samples, including those that fail the visual gate, to preserve diagnostic information. For the official aggregate error, we define the per-query integrity-gated error S_n , assigning a fixed penalty κ to visually invalid rollouts:

$$S_n = \begin{cases} D_n^{\text{NDTW}}, & G_n = 1, \\ \kappa, & G_n = 0. \end{cases} \quad (8)$$

We first average S_n within each task and then report the macro-average across tasks, preventing tasks with more samples from dominating the ranking. Alongside this integrity-gated error, WorldEcho reports the visual-gate pass rate, ungated NDTW error, and results stratified by action-query category. Algorithm 1 summarizes the evaluation procedure.

3.4 WorldSync: Improving Action Following

Taken together, the failures identified above point to an off-expert support gap and weak action dependence in generated dynamics. Closing the former calls for distributional coverage; addressing the latter calls for both representational

Algorithm 1: Integrity-Gated Action-Following Evaluation

Require: AC-WM $\mathcal{W}_\theta$, action queries $\mathcal{Q}$, failure penalty κ

- 1: **for** each query $n: (o_0, c, a_{1:H}, I_{1:H}^{\text{GT}}) \in \mathcal{Q}$ **do**
- 2: Generate $\hat{I}_{1:H} \sim p_\theta(\cdot \mid o_0, c, a_{1:H})$
- 3: Evaluate $G_n = G_{\text{vis}}(\hat{I}_{1:H})$
- 4: Extract $\hat{\tau} = \Phi(\hat{I}_{1:H})$ and $\tau^{\text{GT}} = \Phi(I_{1:H}^{\text{GT}})$
- 5: Compute pose-aware NDTW error D_n^{NDTW}
- 6: Set $S_n \leftarrow D_n^{\text{NDTW}}$ if $G_n = 1$; otherwise $S_n \leftarrow \kappa$
- 7: **end for**
- 8: Aggregate $\{S_n\}$ within each task and then across tasks
- 9: **return** task-macro integrity-gated error, visual pass rate, and ungated NDTW errors

grounding within individual rollouts and intervention-effect alignment across paired rollouts. As illustrated in Figure 4, WorldSync realizes these three requirements through action coverage expansion, an *Action-Forcing Expert* (AFE), and *Intervention-Effect* (IE) supervision, respectively. In short, coverage expansion broadens the action consequences from which the model learns, AFE grounds what its representations encode in robot dynamics, and IE aligns how its predictions change with how the ground-truth futures change.

We train the video backbone with flow matching. Let x_0 denote the clean latent of the target future video and $\epsilon \sim \mathcal{N}(0, I)$ Gaussian noise. At flow time $t \in [0, 1]$, we construct $x_t = (1 - t)x_0 + t\epsilon$ and optimize

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{x_0, \epsilon, t} \left[\|v_\theta(x_t, t \mid o_0, c, a_{1:H}) - (\epsilon - x_0)\|_2^2 \right], \quad (9)$$

where v_θ is the action-conditioned flow velocity predicted by the world model.

Action Coverage Expansion Strategy Expert demonstrations cover only a narrow subset of feasible action consequences, leaving the off-expert support gap identified by our diagnosis. To broaden this support, we train with multi-task simulated trajectories that span expert behavior, local perturbations, cross-state replays, policy rollouts, and broad feasible actions. A small set of target-task real-robot demonstrations is mixed with these simulated data to preserve target-domain visual fidelity. To transfer action-following knowl-

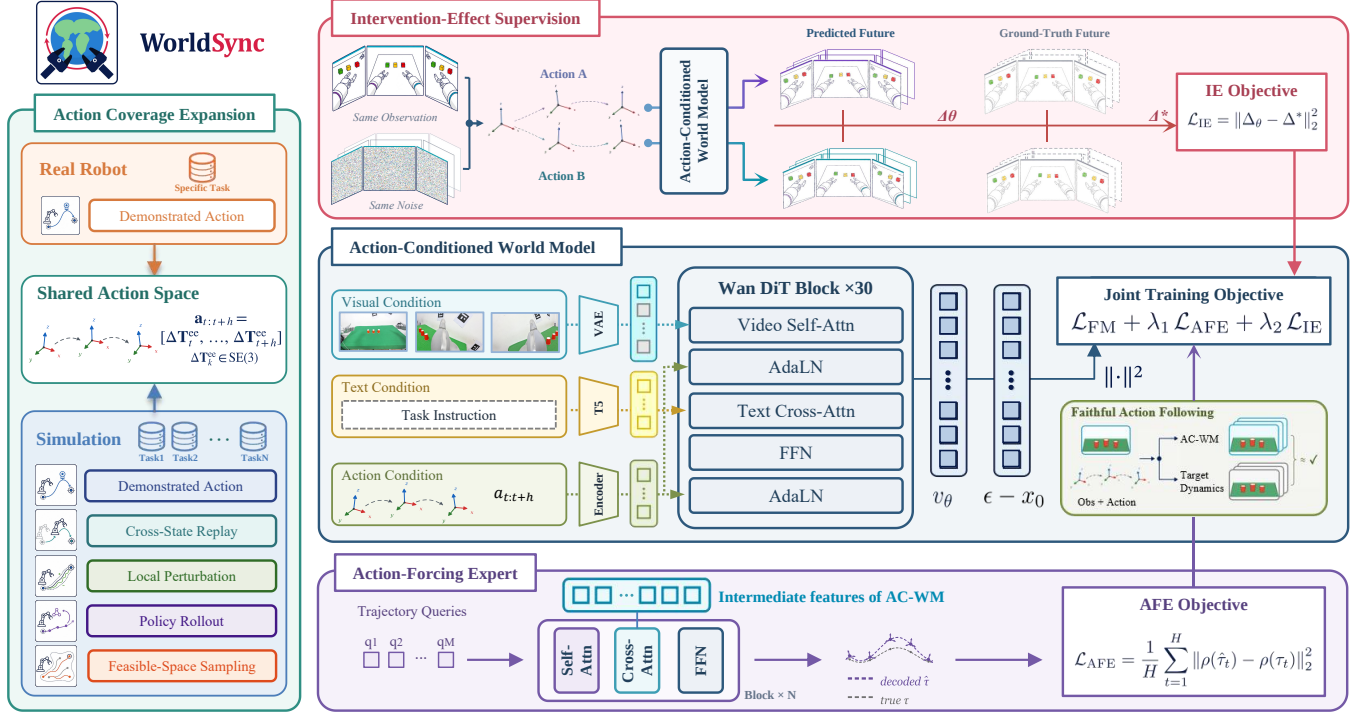


Figure 4: Overview of WorldSync. We expand action coverage by unifying diverse simulated expert and off-expert trajectories with target-domain real-robot demonstrations in a shared SE(3) end-effector action space. The AC-WM generates future videos from visual, language, and action conditions. AFE grounds intermediate video representations in future robot trajectories, while IE supervision aligns predicted and ground-truth intervention effects under shared observations and noise. All objectives are jointly optimized for faithful action following.

edge across the two domains, we represent both simulated and real-robot actions as relative Cartesian end-effector pose displacements expressed in the robot base frame, providing a shared action space for learning relationships between actions and their consequences across simulation and reality.

Action-Forcing Expert Distributional coverage is necessary but does not ensure that intermediate video representations encode the robot dynamics induced by the conditioned actions. To provide an auxiliary feature-level grounding signal, AFE maintains trajectory queries that progressively cross-attend to the intermediate features of successive video blocks and decode the action-induced future end-effector trajectory in SE(3). Given its prediction $\hat{\tau}_{1:H}$ and the ground-truth trajectory $\tau_{1:H}$, we optimize

$$\mathcal{L}_{\text{AFE}} = \frac{1}{H} \sum_{t=1}^H \|\rho(\hat{\tau}_t) - \rho(\tau_t)\|_2^2, \quad (10)$$

where ρ denotes the numerical pose representation used to parameterize translation and orientation. AFE does not directly read the actions or write back to the video stream; its loss instead updates the backbone through the video features. It is removed at inference time.

Intervention-Effect Supervision AFE grounds individual rollouts at the representation level but does not directly supervise how the generated future should change when the

conditioned action changes. IE therefore provides a complementary relational signal using paired trajectories that share the current observation and instruction but execute different actions. Both branches use the same flow-matching noise endpoint, isolating the action as the only differing model input. The predicted and target intervention effects are

$$\Delta_\theta = v_\theta^A - v_\theta^B, \quad \Delta^* = x_0^B - x_0^A, \quad (11)$$

and we align them over future video latents using

$$\mathcal{L}_{\text{IE}} = \|\Delta_\theta - \Delta^*\|_2^2. \quad (12)$$

Thus, beyond fitting each future independently, the model learns how its prediction should change when the conditioned action changes.

Joint Training Objective Combining the standard flow-matching generation loss with the two auxiliary objectives gives

$$\mathcal{L} = \mathcal{L}_{\text{FM}} + \lambda_{\text{AFE}} \mathcal{L}_{\text{AFE}} + \lambda_{\text{IE}} \mathcal{L}_{\text{IE}}. \quad (13)$$

$\mathcal{L}_{\text{AFE}}$ is applied when future trajectory labels are available, while $\mathcal{L}_{\text{IE}}$ is applied to intervention pairs. Together with expanded action coverage, the two auxiliary objectives complement the standard flow-matching objective with representational grounding and intervention-effect alignment for faithful action-conditioned generation.

4 Experiments

We begin by describing the common benchmarks, comparison protocols, and evaluation metrics (§4.1). Building on this protocol, we use WorldEcho to determine whether evaluation on demonstrated actions masks failures under off-expert control (§4.2). We then benchmark WorldSync against six baseline world models and quantify the effect of expanded action coverage (§4.3). To assess downstream utility, we examine whether stronger action following translates into more effective policy improvement under matched budgets (§4.4). Finally, we disentangle the contributions of expanded action coverage, Intervention-Effect supervision, and the Action-Forcing Expert (§4.5).

4.1 Experimental Setup

Benchmarks and evaluation sets. The main evaluation covers 50 RoboTwin manipulation tasks (Mu et al. 2025; Chen et al. 2026b) using the five action-query categories defined by WorldEcho (§4.2, §4.3). Component analysis uses four RoboTwin tasks under the same five-category protocol (§4.5). We separately evaluate policy improvement in RoboTwin and on real robots (§4.4).

Baselines. We compare WorldSync against six baselines spanning complementary robotic world-model paradigms. CtrlWorld (Guo et al. 2026b) serves as a dedicated action-conditioned world model for robot manipulation. Cosmos-Predict2.5 (NVIDIA et al. 2025) and Cosmos3 (NVIDIA 2026) bring large physical-AI foundation models for action-conditioned generation into the comparison, while Dream-Dojo (Gao et al. 2026) provides a generalist robot world model pretrained on large-scale human video. Motus (Bi et al. 2025) and LingBotVA (Li et al. 2026a) further broaden the comparison to unified world-action modeling, using a Mixture-of-Transformers (MoT) architecture and a causal autoregressive formulation, respectively. For each backbone, we evaluate variants trained with either Expert Demonstrations or Expanded Action Coverage on the same task split and action-query set. Table 1 reports their designated endpoints under the common WorldEcho protocol.

Metrics. The primary metric is integrity-gated error. Raw pose-aware NDTW and visual-integrity pass rate separately characterize action mismatch and visual failure. All metrics are macro-averaged over tasks.

4.2 Benchmark Diagnosis

Off-Expert Performance Gap. We evaluated whether demonstrated-action performance reflects behavior under broader feasible control. Across all six expert-trained models, integrity-gated error increased by 0.029–0.099 m on off-expert queries (Figure 5a). Evaluation on demonstrated actions therefore systematically understated errors under feasible but unseen controls.

Failure Decomposition. The gap reflected both trajectory inconsistency and visual degradation. Across models, raw NDTW increased by 0.010–0.043 m and visual failure rate by 6.3–28.1 percentage points; both increases were consistent across all models (Figure 5b). Their relative contributions

varied: some models mainly lost visual integrity, whereas others remained visually plausible but followed the requested motion poorly. Thus, either component metric alone would miss part of the failure.

Expanded Evaluation Coverage. Figure 5c visualizes the distributional coverage of the evaluation queries. Expert actions occupy a relatively compact region of the projected action space, whereas the four off-expert query categories extend evaluation to a much broader region. Thus, WorldEcho evaluates action following over a wider action distribution than expert-only protocols. Together with the failure decomposition above, this broader query distribution exposes two limitations of expert-only AC-WMs: limited support for off-expert action consequences and weak dependence of generated dynamics on the queried actions.

4.3 Main Action-Following Evaluation

Table 1 examines whether Expanded Action Coverage improves action following across different world-model backbones and how the complete WorldSync compares with all baseline configurations under the common WorldEcho protocol.

Effect of Expanded Action Coverage. At their designated endpoints, all six baseline backbones trained with Expanded Action Coverage achieved lower integrity-gated error and raw NDTW than their counterparts trained on Expert Demonstrations. Visual pass rate improved for three backbones, remained nearly unchanged for two, and decreased for one. Expanded Action Coverage therefore consistently strengthened trajectory alignment across architectures, whereas its effect on visual integrity remained backbone-dependent.

Comparison with Baselines. Among all evaluated configurations, WorldSync achieved the lowest integrity-gated error point estimate, slightly lower than CtrlWorld (0.066 versus 0.067), and the highest visual pass rate, slightly exceeding Motus (84.5% versus 84.3%). The component-wise ranking was more nuanced: Cosmos-Predict2.5 attained a lower raw NDTW than WorldSync (0.013 versus 0.022). Thus, WorldSync’s leading integrity-gated result reflects a strong balance between trajectory alignment and visual integrity rather than uniform dominance across individual metrics.

4.4 Policy Improvement

Policy-Improvement Protocol. We adapt VLAW (Guo et al. 2026a) for two matched policy-improvement rounds. Within each domain, we hold the initial policy and the interaction, world-model rollout, and policy-training budgets fixed, varying only the world-model condition. Simulation compares WorldSync with CtrlWorld trained using Expanded Action Coverage or Expert Demonstrations; real-robot evaluation uses the expert-trained CtrlWorld as the baseline.

Simulation Results. From comparable initial success rates of 51–52% on the RoboTwin task, WorldSync reached 65% after two rounds, gaining 13 percentage points (Figure 6). CtrlWorld reached 56% with Expanded Action Coverage and 57% with Expert Demonstrations, gaining 5 points in both cases and finishing 8–9 points behind WorldSync.

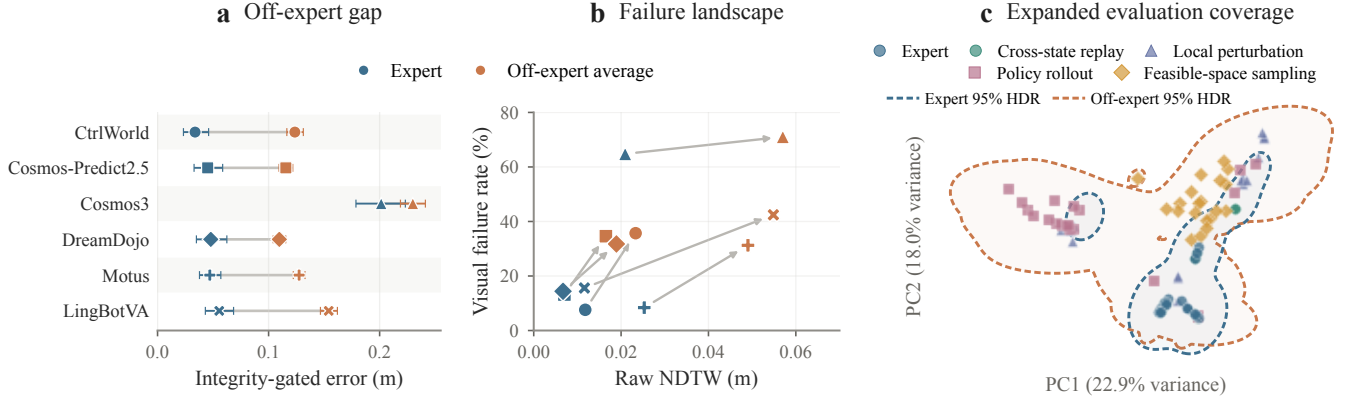


Figure 5: Diagnosing off-expert action following and evaluation coverage. (a) Integrity-gated error on expert and off-expert actions across six world models; error bars show task-bootstrap 95% confidence intervals. (b) Changes in raw NDTW and visual failure rate from expert to off-expert actions. (c) PCA visualization showing that off-expert queries cover a broader action distribution than expert actions.

Table 1: Main WorldEcho comparison on 50 RoboTwin tasks under the frozen evaluation protocol. Baseline models use 20k updates with Expert Demonstrations and 40k updates with Expanded Action Coverage, while WorldSync uses 60k updates. All values are task-macro averages. The best result in each column is shown in bold red, and the second best result is shown in blue.

Model	Training regime	Gated error ↓	Raw NDTW ↓	Visual pass (%) ↑
Baseline world models				
CtrlWorld	Expert Demonstrations	0.0716	0.0266	83.89
CtrlWorld	Expanded Action Coverage	0.0670	0.0210	83.71
Cosmos-Predict2.5	Expert Demonstrations	0.0894	0.0190	75.09
Cosmos-Predict2.5	Expanded Action Coverage	0.0842	0.0127	75.03
Cosmos3	Expert Demonstrations	0.1432	0.0572	63.94
Cosmos3	Expanded Action Coverage	0.1218	0.0419	68.97
DreamDojo	Expert Demonstrations	0.0805	0.0210	78.97
DreamDojo	Expanded Action Coverage	0.0801	0.0151	77.20
Motus	Expert Demonstrations	0.1116	0.0548	75.09
Motus	Expanded Action Coverage	0.0731	0.0292	84.34
LingBotVA	Expert Demonstrations	0.1148	0.0473	71.83
LingBotVA	Expanded Action Coverage	0.0897	0.0340	79.71
WorldSync	Expanded Action Coverage	0.0661	0.0223	84.51

Real-Robot Results. On the real-robot stacking-cups task, both conditions started at 48% success. After two rounds, WorldSync reached 68%, compared with 56% for CtrlWorld, corresponding to gains of 20 and 8 percentage points. In both domains, the complete WorldSync condition combined stronger WorldEcho performance with larger downstream policy gains than the compared CtrlWorld conditions.

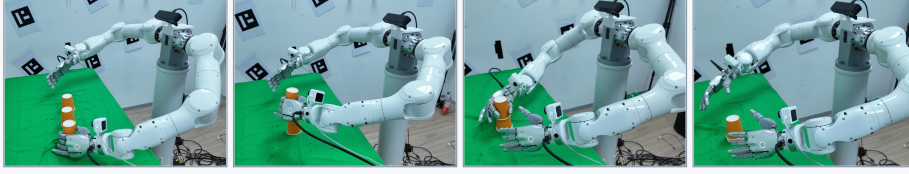
4.5 Component Contributions and Interactions

Table 2: WorldSync ablation of expanded action coverage, intervention-effect (IE) supervision, and the Action-Forcing Expert (AFE) on four RoboTwin tasks. Results are averaged over eight common checkpoints.

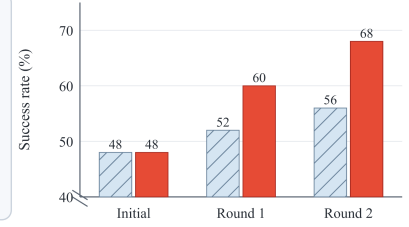
Variant	Coverage	Gated ↓	Raw ↓	Visual (%) ↑
Base	Expert	0.0781	0.0306	82.41
Base	Expanded	0.0738	0.0258	82.68
+ IE	Expanded	0.0700	0.0170	81.25
+ AFE	Expanded	0.0753	0.0284	83.04
Full	Expanded	0.0695	0.0189	81.96

Expanded Action Coverage. With IE and AFE disabled, expanding the training coverage reduced mean gated er-

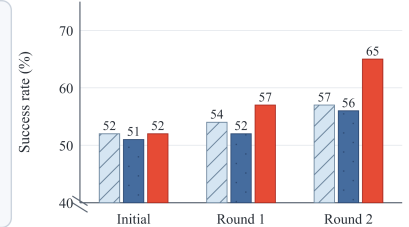
Real robot: stacking cups



Policy improvement



Simulation: dump bin bigbin



WorldSync CtrlWorld with Expanded Action Coverage CtrlWorld with Expert Demonstrations

Figure 6: Policy improvement under matched budgets on a RoboTwin bin-dumping task and a real-robot stacking-cups task. Success rates are reported for the initial policies and after each of two refinement rounds.

ror from 0.0781 to 0.0738 and raw NDTW from 0.0306 to 0.0258, while the visual pass rate remained nearly unchanged. This isolates broader coverage as a source of improved action consistency rather than visual-quality gains.

Roles and Interaction of IE and AFE. Under expanded action coverage, IE produced the main trajectory gains and achieved the lowest raw NDTW of 0.0170. AFE alone improved neither action metric, although it yielded the highest visual pass rate. Adding AFE to IE slightly lowered gated error and partially recovered the visual pass rate relative to IE alone, yielding the best gated result for the full model at 0.0695. These comparisons identify IE as the primary driver of trajectory alignment, whereas AFE contributes conditionally by improving the balance between action consistency and visual validity.

5 Conclusion and Limitations

In this work, we introduced WorldEcho to evaluate action-conditioned world models beyond the narrow distribution of expert demonstrations, jointly measuring visual integrity and action-induced trajectory alignment. Our evaluation reveals that the evaluated models consistently degrade under feasible off-expert actions, exposing failures overlooked by expert-only protocols. Guided by this diagnosis, we proposed WorldSync, which combines distributional coverage, representational grounding, and intervention-effect alignment for faithful action-conditioned generation. Across RoboTwin and real-robot experiments, these improvements produced more reliable world-model rollouts and translated into greater gains during iterative policy improvement. Although WorldEcho substantially broadens evaluation coverage, comprehensively probing long-horizon interactions across diverse embodiments and open-world environments remains a shared challenge for the field and an important

direction for future work.

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